MIL-R-81113B(AS)
20 November 1972
SUPERSEDING
MIL-R-81113A(AS)
15 August 1970

MILITARY SPECIFICATION

RADAR TRANSPONDER
AN/DPN-78(*)

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 Scope - The equipment covered by this specification shall receive interrogations from "X" band radars and transmit a reply to the interrogation on another frequency within the "X" band. The radar transponder is used to extend the operating range of the radar.

1.1.1 Basic or prime equipment - The equipment covered by this specification shall consist of the following items:

Item	Type Designation	Appl. Para.
Receiver-Transmitter	RT-(*)	3.5.1

1.1.2 Associated equipment - The associated equipment shall consist of:

a. One (1) Antenna AT/(*)

b. One (1) Test Set Shop, AN/UPM-(*)

c. One (1) Test Set, Flight Line AN/DPM-(*)

d. One (1) Set of Special Support Equipment (See para. 6.4)

e. One (1) Set of Inter-connecting Cables

1.2 Classification - The equipment covered by this specification is classified as production equipment.

FSC 5895

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2. [illegible]CABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-172	Cases; Bases, Mounting, and Mounts
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-E-17555	Electronic and Electrical Equipment, Accessories, and Repair Parts: Packaging and Packing of
MIL-T-18303	Test Procedures; Preproduction and Inspection, for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment

Naval Air Systems Command

AR-10	Maintainability of Avionics Requirement and Systems, General Requirements for

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-461	Electromagnetic Interference Characteristics, Requirements for Equipment

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STANDARDS

Military (Cont'd)

MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-471	Maintainability Demonstration
MIL-STD-480	Configuration Control, Engineering Changes, Deviations and Waivers
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-781	Reliability Tests Exponential Distribution

(When requesting specifications, standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.)

3. REQUIREMENTS

3.1 Qualification - The transponders furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.3 and 6.7).

3.2 Parts and materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing the following shall govern:

(1) Parts and materials requirements shall conform to MIL-E-5400.

(2) Nonrepairable subassemblies, as outlined in MIL-E-5400, shall be used when practicable. The general size of the subassembly, and the amount of circuitry to be included therein shall be approved by the procuring activity. Nonrepairable subassemblies must be reliable. A mean-time-to-failure of more than 5000 hours should be the goal of each.

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(3) When previously produced models of the equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies, without the approval of the procuring activity.

3.2.1 Nonstandard parts and material approval - Approval for the use of nonstandard parts and materials shall be obtained as outlined in MIL-E-5400.

3.3 Design and construction -

3.3.1 Design data - No data is required by this specification or by applicable documents referenced in Section 2 unless specified in the contract or order.

3.3.1.1 General - The equipment shall conform with all the applicable requirements of MIL-E-5400 for design, construction and workmanship, unless otherwise specified herein. Engineering changes, waivers and deviations shall be in accordance with MIL-STD-480.

3.3.2 Total weight - The total weight of the equipment, excluding cables, shall be a minimum consistent with good design and shall not exceed 6.0 pounds.

3.3.3 Operating Life -

3.3.3.1 Operational stability - The equipment shall operate with optimum performance for 25 hours, continuously or intermittently, without readjustment.

3.3.3.2 Total operating life - The equipment shall have a minimum total operating life of 1000 hours with reasonable servicing. Replacement of parts shall be specified by the manufacturer.

3.3.3.3 Mean-time-between-failures - The equipment shall have a mean-time-between-failures of at least 250 hours. A failure shall be defined as any malfunctioning which causes the equipment to fail to meet the minimum performance requirements specified for the equipment.

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3.3.4 Cabling and connections -

3.3.4.1 Cables and connectors - The equipment shall provide for the use of cables and connectors in accordance with MIL-E-5400. Plugs shall be provided to mate with all external connectors.

3.3.4.2 Input power connection - Input power connections shall be in accordance with paragraph 3.5.1.6.

3.3.5 Interchangeability - The equipment shall meet the interchangeability requirements as defined in MIL-E-5400.

3.3.6 Interference control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of MIL-STD-461.

3.3.7 Maintainability provisions - The equipment shall meet the maintainability requirements of AR-10.

3.3.8 Nomenclature and nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with MIL-N-18307.

3.3.8.1 Type designation - The parentheses (*), when used in the type designation, shall be replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with paragraph 3.3.8. The complete type number shall be used on nameplates, shipping records, and instruction books as applicable.

3.3.9 Standard conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room ambient (30° ± 10°C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90% RH
Input power voltage	27.5 ± 0.5 VDC

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3.3.10 Service conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in MIL-E-5400 for Class 2 equipment, except as modified herein.

3.3.10.1 Vibration - The equipment shall operate satisfactorily when subjected to the vibration requirements of Curve IV of MIL-E-5400 with the frequency limit extended to 2000 cps.

3.3.10.2 Shock - The equipment shall operate satisfactorily when subjected to a shock of 100 G (11 millisecond duration), on any axis.

3.3.10.3 Acceleration - The equipment shall operate satisfactorily when subjected to an acceleration of 50 G, on any axis.

3.3.10.4 Pressurization - The entire equipment shall be pressurized in accordance with MIL-E-5400N, paragraph 3.2.21.

3.3.11 Warm-up time - The time required for the equipment to warm-up prior to operations shall be kept to a minimum and shall not exceed 3 minutes under standard conditions and 5 minutes at extreme service conditions.

3.3.12 Primary input power requirements - The equipment sheet meet all applicable requirements of MIL-STD-704 and shall give specified performance from the following power sources with characteristics as defined in MIL-STD-704 having limits as specified therein, except that the maximum surge voltage on all test curves be limited to 70 volts.

(1) 28 V nominal DC power ranging from 21 to 31 VDC. 1.55 amperes maximum current at a pulse repetition frequency of 2000 Hz.

3.3.12.1 Undervoltage protection - The equipment shall not be damaged by voltages below the minimum specified herein and shall automatically resume normal operation when the voltage returns within limits.

3.3.13 Compatibility - The form factor of the equipment shall be compatible with that specified in paragraph 3.5. A reduction in size is permissible provided an adapter plate is supplied to match the mounting provisions, and that the connectors as specified are used.

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3.4 Performance -

3.4.1 General - Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under both standard and extreme service conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified.

3.4.2 Frequency characteristics - The equipment shall operate in the "X" band of frequencies as follows:

3.4.2.1 Receiver - The center of the receiver pass band shall be adjustable to any frequency from 9100 to 9600 MHz.

3.4.2.1.1 Factory tuning - Unless otherwise specified in the contract, the equipment shall be delivered with the receiver tuned to 9250 MHz.

3.4.2.2 Transmitter - The transmitter frequency shall be adjustable to any frequency from 9200 to 9500 MHz.

3.4.2.2.1 Factory tuning - Unless otherwise specified in the applicable contract, the equipment shall be delivered with the transmitter tuned to 9350 MHz.

3.4.3 Receiver characteristics -

3.4.3.1 Interrogation signal - The equipment shall receive radar interrogation signals. The interrogation signal may be either single or paired pulses as specified in paragraph 3.4.5. The choice of the double pulse signal shall be accomplished by internal soldering of a jumper wire.

3.4.3.2 Trigger sensitivity and stability - The equipment shall have a trigger sensitivity of at least -70 dbm over a frequency range of ±2 MHz from the nominal frequency. Full triggering shall be defined as at least 99% replies.

3.4.4 Selectivity -

3.4.4.1 Bandwidth - The equipment shall have a bandwidth of 11 ± 3 MHz at the 3 db points.

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3.4.4.2 Image - The image frequency shall be at least 60 db of signal rejection.

3.4.5 Decoder characteristics -

3.4.5.1 Single pulse interrogation - The equipment shall respond to a single pulse interrogation having the following characteristics:

(1)	Pulse width	0.2 to 0.6 microsecond
(2)	Pulse rise time	0.1 microsecond or less

3.4.5.2 Double pulse interrogation - When modified for this mode of operation, the equipment shall respond to a double pulse interrogation having the following characteristics:

(1)	Pulse space	1.5 to 12.4 microseconds continuously adjustable
(2)	Pulse width	0.2 to 0.6 microsecond
(3)	Pulse rise time	0.1 microsecond or less

3.4.5.2.1 Double pulse reject limit - Change to read: The equipment shall be capable of rejecting interrogations having a spacing of ±.2 microseconds of the chosen spacing. The reject limits will be adjustable from ±.15 microsecond to ±1.0 microsecond.

3.4.6 Transmitter characteristics -

3.4.6.1 Frequency stability - After a 3 minute warm-up, the transmitter frequency shall be within ±2 MHz of the nominal frequency under standard conditions. After a 15 minute warm-up, the transmitter frequency shall not drift more than 0.05 MHz/degree C. temperature change.

3.4.6.2 Reply signal characteristics - The equipment shall reply (to properly coded interrogations) with a single pulse having the following characteristics:

(1)	Pulse width	0.25 ± 0.1 microseconds
(2)	Pulse rise time	0.1 microsecond or less

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(3)	Leading edge jitter	±0.05 microsecond at an input signal level of -65 dbm

3.4.6.3 Transmitter power - The peak pulse power from the transmitter antenna terminal shall be at least 300 watts at 2000 PRF.

3.4.6.4 Overinterrogation protection - The radar beacon shall not suffer permanent damage when subjected to interrogations which would result in exceeding the duty cycle limitation of 0.002. This circuitry may utilize the recovery time of the beacon provided there is no degradation in performance from that specified in paragraph 3.4.9.

3.4.7 Beacon delay time - The fixed delay time between the single interrogation pulse or the second pulse of the double pulse interrogation and the beacon reply pulse shall be adjustable within the limits of 1.5 to 9.0 microseconds.

3.4.7.1 Delay variation with signal level - The total variation in delay shall not exceed ±0.05 microseconds for signal levels or 0 dbm to -65 dbm.

3.4.8 Random triggering - The transponder shall not trigger due to random noise when the receiver RF input is terminated with a dummy load.

3.4.9 Beacon recovery time - The beacon recovery time shall not exceed 50 microseconds and the receiver sensitivity shall have returned to at least -70 dbm 50 microseconds after the receipt of an acceptable interrogation signal.

3.4.10 Antenna terminal - The receiver and transmitter shall have one antenna connection.

3.4.11 Maximum signal input - The transponder shall not be damaged by input signal levels of +10 dbm or less.

3.4.12 Antenna system compatibility - The equipment shall operate satisfactorily in conjunction with an antenna system having a VSWR of 2:1 or better.

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3.4.13 Impedance - The nominal input and output impedance of the radar beacon shall be 50 ohms.

3.4.14 Receiver protection - No damage shall result from operation at 50 MHz separation between the transmitter and receiver.

3.5 Detail requirements -

3.5.1 Receiver-transmitter RT-(*) - The receiver-transmitter unit shall meet the following requirements.

3.5.1.1 Function - The receiver-transmitter receives interrogations from "X" band radars operating on the same frequency that the beacon receiver is tuned to; the interrogation signal is decoded and is used to trigger the beacon transmitter that is tuned to another frequency within the "X" band.

3.5.1.2 Form factor - The unit shall meet the materials, bonding, shielding, and performance requirements of MIL-C-172. The mounting holes are shown in Figure 1.

3.5.1.3 Configuration - The unit shall consist of a center body to which all the internal parts and the connectors are attached, a removable top cover and a removable bottom plate. The internal arrangement of components shall be such as to provide good accessibility for maintenance and testing.

3.5.1.4 Contents of unit - The receiver-transmitter unit shall comprise subassemblies that perform the following functions:

(1) Preselector

(2) Receiver local oscillator

(3) Receiver mixer

(4) Receiver IF amplifier

(5) Video amplifier

(6) Decoder

(7) Modulator

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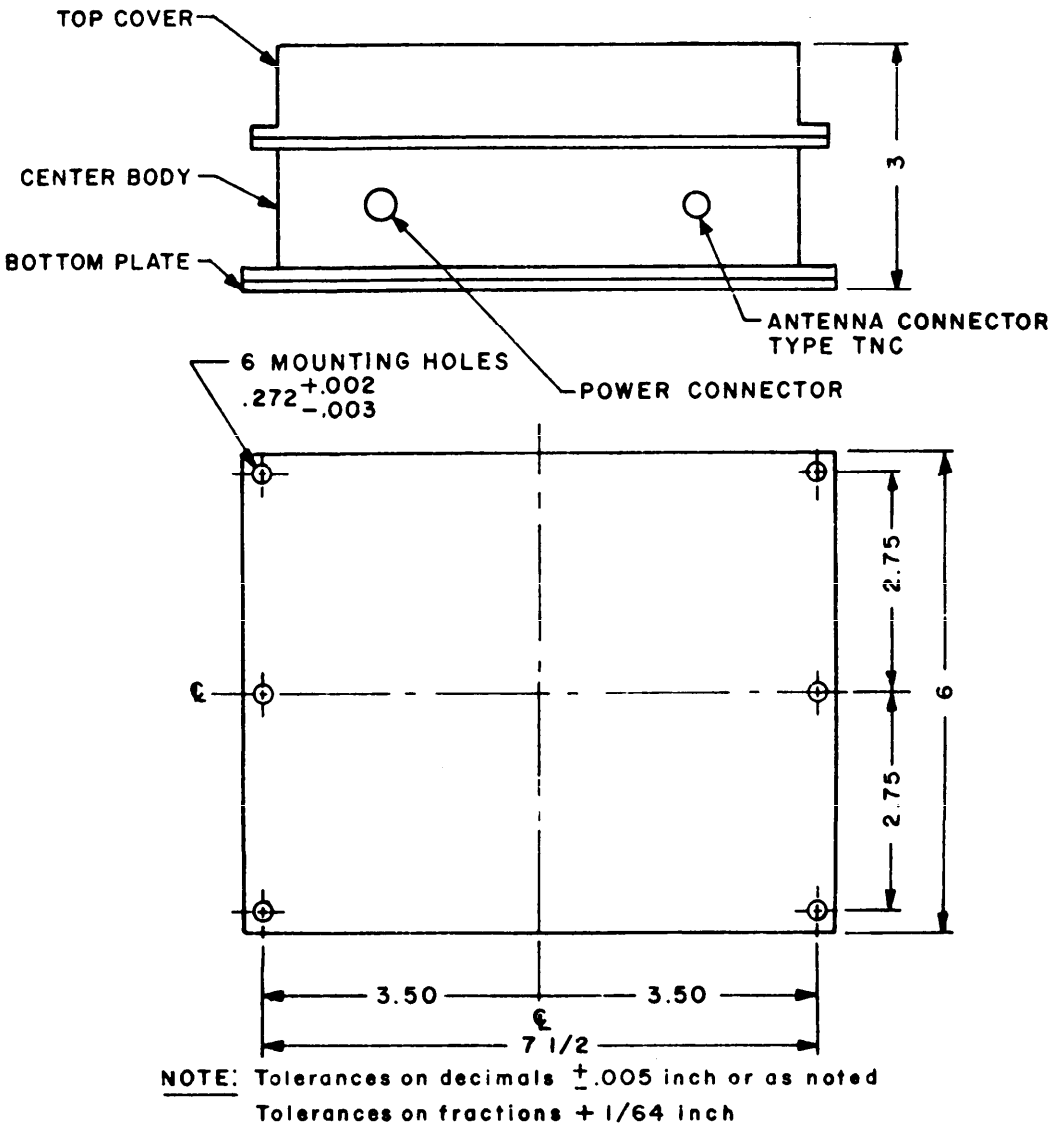


Figure 1. Radar Transponder

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(8) Magnetron

(9) Four port circulator

(10) Power supply

3.5.1.5 Controls - The following controls shall be provided internal to the receiver-transmitter. All adjustments shall be accessible from the top with the top cover removed.

(1) Magnetron Frequency Control

(2) Receiver Local Oscillator Frequency Control

(3) Trigger Level Control

(4) Paired Pulse Adjustment

(5) Preselector Tuning

(6) Accept/Reject Adjustment

(7) Overinterrogation Protection Adjustment

(8) Blanking Adjustment

(9) Beacon Delay Adjustment

3.5.1.6 Electrical connections - Connections to external circuits shall be provided by standard receptacles in accordance with MIL-E-5400 as follows:

Reference Designation	Receptacle Type	Function
J1	PT07C-10-6P	Power Inlet (Pin A Positive; Pin D negative)
J2	TNC	Antenna

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections - The inspection requirements specified herein are classified as follows:

1. Qualification inspection (see 4.3)
2. Quality conformance inspection (see 4.4)
3. Inspection of preparation for delivery (see 4.5)

4.3 Qualification inspection - Qualification test samples shall be subjected to all examinations and tests as indicated in Table I.

4.3.1 Qualification inspection sample size - The qualification inspection sample shall consist of three transponders separately packaged and forwarded to the activity responsible for qualification designated in the letter of authorization. The samples shall in all respects be representative of the manufacturer's production item. Samples shall be plainly identified by securely attached durable tags marked with the following information:

Sample for Qualification Test
Specification MIL-R-81113B(AS)
Radar Transponder AN/DPN-78
Manufacturer's Part Number
Name of Manufacturer
Submitted (date) under authorization
(reference letter authorizing the test)

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TABLE I. QUALIFICATION INSPECTIONS

Examination or Test	Requirement Paragraph	Examination or Test Pargraph
Examination of product	3.2.3.3	4.6.1
Operational test	3.4	4.6.2
Environmental tests	3.3.10	4.6.3
Interference test	3.3.6	4.6.4
Reliability test	3.3.3	4.6.5
Maintainability test	3.3.7	4.6.6
Time to change operating frequency	-	4.6.8
Flight test	-	4.6.9

4.3.2 Retention of qualification - In order to retain qualification, whenever a design or material change is made tests shall be conducted to determine the affect of such a change on the performance of the equipment and to assure adequate quality control. Such tests shall be made on an early equipment after an engineering or material change. Tests shall also be conducted whenever failure reports or other information indicate that additional tests are required. (This will be determined by the procuring activity.) Periodic verification shall be by certification at intervals of no more than two years.

4.3.2.1 Scope of tests - Tests shall consist of such tests as approved by the procuring activity. The contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

4.3.2.2 Test report - The contractor shall furnish the activity responsible for qualification with a certified test report, in duplicate, showing the results of the tests.

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4.4 Quality conformance inspection - Quality conformance inspection shall consist of the following:

(a) Individual inspections (see 4.4.1)

(b) Lot acceptance inspections (see 4.4.2)

4.4.1 Individual inspections - Each transponder shall be subjected to the following inspections or tests:

(a) Examination of product (see 4.6.1)

(b) Operational test (see 4.6.2)

(c) All equipment test (see 4.6.7)

These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability.

4.4.2 Lot acceptance inspections - Lot acceptance inspections may be conducted by the Government on transponders which have successfully passed the individual tests. Equipments shall be selected for lot acceptance inspection by the Government and shipped on Government Bills of Lading to a designated Government activity for tests. The lot sampled for lot acceptance inspection will be placed in a secure storage area by the contractor at his facility pending determination on lot acceptance. The lot sampled will not be packaged for shipment until a determination on lot acceptance is made or after the expiration of 30 days from the date the sample is taken.

4.4.2.1 Lot formation - A lot shall consist of the quantity of equipment produced and presented for acceptance in one month. Two equipments shall be selected at random from the lot and subjected to this test.

4.4.2.2 Scope of tests - As a minimum, each equipment selected for lot acceptance inspection shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

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(2) Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction, production and special tests.

(3) Reliability Assurance test specified in 4.6.5.

4.4.3 Test procedures - The procedures used for conducting quality conformance inspection shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity of the Government Inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts, such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary.

4.4.4 Equipment failure - Should a failure occur during any of the quality conformance inspections, the following action shall be taken:

(a) Determine the cause of failure.

(b) Determine if the failure is a material defect or a design defect.

(c) Submit to the procuring activity for approval, proposed corrective action to reduce the possibility of the same failure(s) occurring in future tests.

(d) Where practical, include a test in the individual test to check all equipment for this condition until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Inspection of preparation for delivery - Sample packages shall be selected in accordance with MIL-STD-105, Inspection Level S-3, AQL 2.5%, and inspected for preservation, packaging, packing, and marking for shipment and storage in accordance with the requirements of Section 5.

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4.6 Test methods -

4.6.1 Examination of product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.6.2 Operational test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.6.3 Environmental tests - Environmental tests shall be in accordance with MIL-T-5422.

4.6.4 Interference tests - Interference tests shall be in accordance with MIL-STD-461 and MIL-STD-462.

4.6.5 Reliability test - The reliability test shall be in accordance with MIL-STD-781. Test Plan XXVIII, Test Level E except as specified herein. Three (3) equipments shall be subjected to this test. Each equipment shall be operated for 1000 hours and the total cumulative time shall be 3000 hours. The length of the Heating Period shall be one (1) hour. The equipment shall be ON during the Heating Period and OFF during the Cooling Period. Measurements of performance (see 3.4) shall be made at least once during each Heating Period. This test shall have been completed satisfactorily if no more than 12 relevant failures are experienced during the 3000 hours cumulative time. The corrective maintenance time associated with each failure experienced during this test shall be measured and recorded for use in the Maintainability Test (see 4.6.6). Equipment subjected to this test may be restored and delivered as items under the contract provided they pass quality conformance inspection (see 4.4).

4.6.6 Maintainability test -

4.6.6.1 Corrective maintenance - A minimum of 47 failures shall be corrected to determine the corrective maintenance time distribution. Except for the failures that occur during the Reliability Test and whose corrective maintenance time is included in the data for this test, the failures to be corrected shall be obtained by simulation. Fault simulation for a corrective maintenance task shall be obtained by introducing a faulty part to the equipment. Only one fault shall be introduced to the equipment for each corrective maintenance task. Corrective maintenance tasks to be used in the test shall be

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selected in accordance with Appendix A of MIL-STD-471. The test shall be performed by personnel with skills equivalent to a Navy Electronic Technician 3rd Class using test equipment as specified in the equipment handbook. The time elements of fault isolation, disassembly, interchange, reassembly, alignment and checkout shall be measured and recorded for each corrective maintenance task time (see 6.5.7) during the test. This test shall have been completed satisfactorily if the arithmetic average time to perform the 47 corrective maintenance tasks is equal to or less than 31 minutes and if no more than four (4) corrective maintenance tasks are performed in 52 minutes or more.

4.6.7 All equipment test - Each equipment shall be tested for 25 consecutive failure free hours of operating time in accordance with MIL-STD-781, Test Plan XXIX, Test Level E except as specified herein. The length of the Heating Period shall be one (1) hour. The equipment shall be ON during the Heating Period and OFF during the Cooling Period. Measurements of performance (see 3.4) shall be made at least once during each Heating Period. To determine whether or not the MTBF is being met at any time, the operating test hours and the failures occurring therein shall be recorded similar to the example in 6.6 and compared with the values listed in Table II. The test record, similar to the example in 6.6, shall be kept current so that the experience from the beginning of the contract is included. At the conclusion of each month the test results shall be sent to the procuring activity and to the Naval Air Systems Command, Attn: AIR-53523. At any time that the cumulative operating time to failure for the most current six (6) failures is less than 788 hours, the procuring activity reserves the right to stop the acceptance of equipment at any time that the cumulative operating time to failure for the most current six (6) failures is less than 738 hours pending a review of the contractor's efforts to improve the equipment. Within 30 days of the notification required above, the contractor shall submit a recommended course of corrective action to the procuring activity.

TABLE II

Failure Number	Data Base (Failures)	Cumulative Test Time (Hours)
1	1	26.4
2	2	133.0
3	3	275.5

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TABLE II (Cont'd)

Failure Number	Data Base (Failures)	Cumulative Test Time (Hours)
4	4	436.3
5	5	608.3
6 and subsequent	6	788.0

4.6.8 Time to change operating frequency - A minimum of ten (10) operating frequency changes shall be performed to determine the average time to change operating frequency to any frequency in the operating range. One crystal shall be interchanged for each operating frequency change. The time elements of disassembly, interchange, reassembly, alignment, and checkout shall be measured and recorded for each operating frequency change. This test shall be performed using the procedures and test equipment specified in the equipment handbook. This test shall have been completed satisfactorily if the arithmetic average time to perform the 10 operating frequency changes is equal to or less than 42 minutes.

4.6.9 Flight test - One equipment shall be subjected to flight tests in targets to determine compliance with all applicable design and performance requirements under field conditions.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with specification MIL-E-17555.

6. NOTES

6.1 Intended use - The equipment specified herein will be installed in an airborne target.

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6.2 Ordering data - Procurement documents should specify the following:

(a) Title, number and date of this specification

(b) Levels of packaging and packing of shipments required (see 5.1)

6.3 Test values - Normal and limiting values of performance data shall be determined at input voltages of 27.5 ± 0.5 VDC. These data are to be used in testing the equipment at installation points for compliance with minimum acceptable standard of performance.

6.4 Special support equipment - The special support equipment or its equivalent recommended to maintain this equipment is listed as follows:

Signal Generator	HP620A
Frequency Meter	PRD 559A
Detector Mount	HP420A
Oscilloscope	Tektronix 547 & Type H Plug-In
Pulse Generator	VPL 715
Directional Coupler	PRDX4420A-20
Waveguide-Coax Adapter	HPH 281A
Power Meter	HP431C
Thermistor Mount	HP478B
Volt-ohm Meter	Simpson 260
50-ohm Load	Tektronix 011-0049-01

6.5 Definitions - The following definitions shall apply:

6.5.1 Interrogation - An interrogation is one or two radio frequency pulses transmitted by an interrogator or radar set.

6.5.2 Reply - A reply is a radio frequency pulse transmitted by a beacon as a result of an interrogation.

6.5.3 Pulse width - Pulse width is that interval of time between points at which amplitude of the pulse is 50 percent of the pulse amplitude.

6.5.4 Pulse amplitude - Pulse amplitude is the maximum voltage amplitude of the pulse of RF pulse envelope, excluding spike.

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6.5.5 Pulse rise time - Pulse rise time is that interval of time required for the leading edge of the pulse to rise from 10 to 90 percent of the pulse amplitude.

6.5.6 Pulse spacing - Pulse spacing is that interval of time between 50 percent pulse amplitude points on the leading edges of the pulse involved.

6.5.7 Corrective maintenance task time - The elapsed time needed to perform a corrective maintenance task consists of:

(a) Fault isolation - Determining, by the use of accessory test equipment and built-in test points, the location of a failure to the extent necessary to effect repair.

(b) Disassembly - Opening the equipment and taking it apart to the extent necessary to make the failed part accessible for removal. This does not include the actual removal time.

(c) Interchange - Removing the failed part and installing a new, interchangeable part.

(d) Reassembly - Putting the equipment back together and closing it.

(e) Alignment - Performing the adjustments that are necessary to return the equipment to satisfactory functioning condition.

(f) Checkout - Verifying that the equipment has been returned to satisfactory functioning condition.

6.5.8 Relevant failure - A relevant failure is any failure or non-specification performance occurring in or exhibited by an item of hardware undergoing test which has not been induced by human error or by failures occurring in or malfunctioning of associated, internal or external equipment or devices.

6.6 Example of all equipment test record - See attachments.

6.7 Qualification - With respect to products requiring qualification, awards will be made only for products which are, at the time

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set for opening of bids, qualified for inclusion in the applicable Qualified Products List (QPL), whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification.

6.7.1 Application for qualification - The activity responsible for the QPL for MIL-R-81113 is the Naval Air Systems Command. The Naval Avionics Facility, Indianapolis, Indiana, has been designated by the Naval Air Systems Command as agent for the establishment of the QPL. Requests for information pertaining to and applications for qualification should be addressed to:

Commander
Naval Air Systems Command
Washington, D.C. 20360
Attn: AIR-53355

Preparing Activity
Navy - AS
(Project No. 5895-N196)

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ALL EQUIPMENT TEST RECORD

Test #	Equip-ment #	Test Time (T)	Cumulative Test Time (Hrs)	Test Date Started/Completed	Remarks
1	1	25	25.0		Accept
2	2	25	50.0		Accept
3	3	25	75.0		Accept
4	4	25	100.0		Accept
5	5	12.1	112.1		Repair and retest. $\Sigma T > 26.4$ hrs @ 1st failure
6	5	25	137.1		Accept
7	6	25	162.1		Accept
8	7	25	187.1		Accept
9	8	25	212.1		Accept
10	9	25	237.1		Accept
11	10	25	262.1		Accept
12	11	25	287.1		Accept
13	12	25	312.1		Accept
14	13	25	337.1		Accept
15	14	25	362.1		Accept
16	15	25	387.1		Accept
17	16	25	412.1		Accept

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ALL EQUIPMENT TEST RECORD (Cont'd)

Test #	Equip-ment #	Test Time (T)	Cumulative Test Time (Hrs)	Test Date Started/Completed	Remarks
18	17	25	437.1		Accept
19	18	25	462.1		Accept
20	19	25	487.1		Accept
21	20	25	512.1		Accept
22	21	6.7	518.8		Repair and retest. $\Sigma T > 133$ hrs @ 2nd failure
23	21	25	543.8		Accept
24	22	25	568.8		Accept
25	23	25	593.8		Accept
26	24	10.1	603.9		Repair and retest. $\Sigma T > 275.5$ hrs @ 3rd failure
27	24	25	628.9		Accept
28	25	25	628.9		Accept
29	26	25	653.9		Accept
30	27	25	703.9		Accept
31	28	25	728.9		Accept
32	29	25	753.9		Accept

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ALL EQUIPMENT TEST RECORD (Cont'd)

Test #	Equip-ment #	Test Time (T)	Cumulative Test Time (Hrs)	Test Date Started/Completed	Remarks
33	30	25	778.9		Accept
34	31	25	803.9		Accept
35	32	25	828.9		Accept
36	33	25	853.9		Accept
37	34	25	878.9		Accept
38	35	25	903.9		Accept
39	36	25	928.9		Accept
40	37	25	953.9		Accept
41	38	25	978.9		Accept
42	39	25	1003.9		Accept
43	40	25	1028.9		Accept
44	41	25	1053.9		Accept
45	42	25	1078.9		Accept
46	43	16.1	1095.0		Repair and retest $\Sigma T > 436$ hrs. @ 4th failure
47	43	25	1120.0		Accept
48	44	25	1145.0		Accept
49	45	25	1170.0		Accept

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ALL EQUIPMENT TEST RECORD (Cont'd)

Test #	Equip-ment #	Test Time (T)	Cumulative Test Time (Hrs)	Test Date Started/Completed	Remarks
50	46	251	1195.0		Accept
51	47	25	1120.0		Accept
52	48	25	1245.0		Accept
53	49	25	1270.0		Accept
54	50	25	1295.0		Accept
55	51	25	1320.0		Accept
56	52	10.4	1330.4		Repair and retest $\Sigma T > 608.3$ hrs. @ 5th failure
57	52	25	1355.4		Accept
58	53	25	1380.4		Accept
59	54	25	1405.4		Accept
60	55	25	1430.4		Accept
61	56	25	1455.4		Accept
62	57	25	1480.4		Accept
63	58	25	1505.4		Accept
64	59	25	1530.4		Accept
65	60	25	1555.4		Accept
66	61	25	1580.4		Accept

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VARIABLE TRANSFORMER (SEE TABLE III)

ALL EQUIPMENT TEST RECORD (Cont'd)

Test #	Equip-ment #	Test Time (T)	Cumulative Test Time (Hrs)	Test Date Started/Completed	Remarks
67	62	25	1605.4		Accept
68	63	11.9	1617.4		Repair and retest $\Sigma T >$ 788 hrs @ 6th failure
69	63	25	1642.3		Accept
70	64	25	1667.3		Accept
71	65	25	1692.3		Accept
72	66	3.8	1696.1		Repair and retest $\Sigma T_7 - \Sigma T_1$ = 1584 788 hrs. @ 7th failure

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OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

MIL-R-81113B(AS) RADAR TRANSPONDER AN/DPN-78(*)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

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